

# Decision Making with Sentiment Analysis in R

From Social Media Noise to Managerial Action

Mustafa Asım Ruhi 401190044  
ruhi.mustafa@stud.hs-fresenius.de

Fresenius University of Applied Sciences  
Course: Data Analysis for Decision-Making  
Professor: Prof. Dr. Stephan Huber

2 June 2026

## **Research Question**

**How can customer tweets be transformed into decision-relevant insights for airline management?**

# What is Sentiment Analysis?

Sentiment analysis is a text mining method used to identify opinions, emotions, and attitudes in written language.

**Positive**

**Neutral**

**Negative**

## Why It Matters for Decision-Making

Companies receive large amounts of customer feedback.

Managers need fast and structured insights.

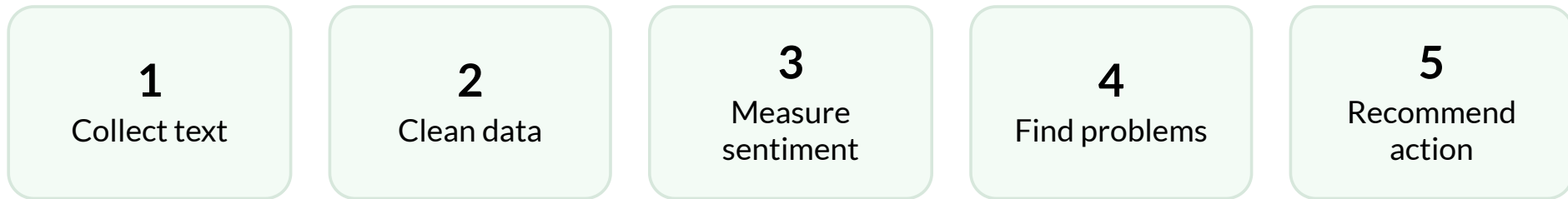
Sentiment analysis helps prioritize business actions.

# Different Sentiment Analysis Methods

| Method            | Basic idea                                 | Why not used / selected?                                            |
|-------------------|--------------------------------------------|---------------------------------------------------------------------|
| Lexicon-based     | Uses predefined sentiment dictionaries     | Selected because it is transparent and easy to reproduce in RStudio |
| Machine learning  | Trains a model on labeled text data        | Requires training/testing setup                                     |
| Transformer-based | Uses context-aware models such as BERT     | More complex for a short live demo                                  |
| Hybrid            | Combines dictionaries and machine learning | More advanced setup                                                 |

In this project, I use a lexicon-based approach because it is transparent, reproducible, and suitable for a short RStudio demonstration.

# From Text to Managerial Action

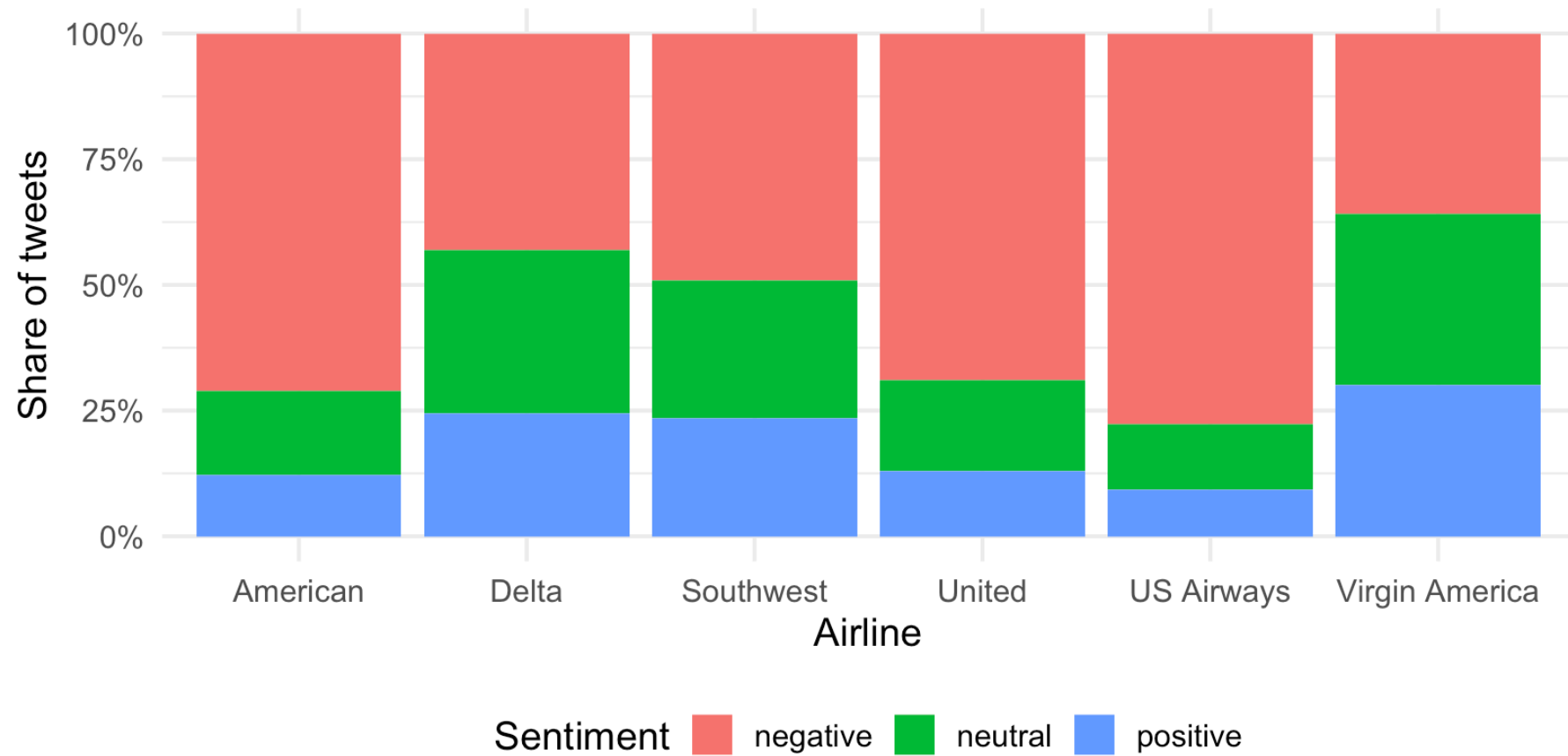


## Case Study: Airline Tweets

The example uses **14,640** airline-related tweets classified as negative, neutral, or positive.

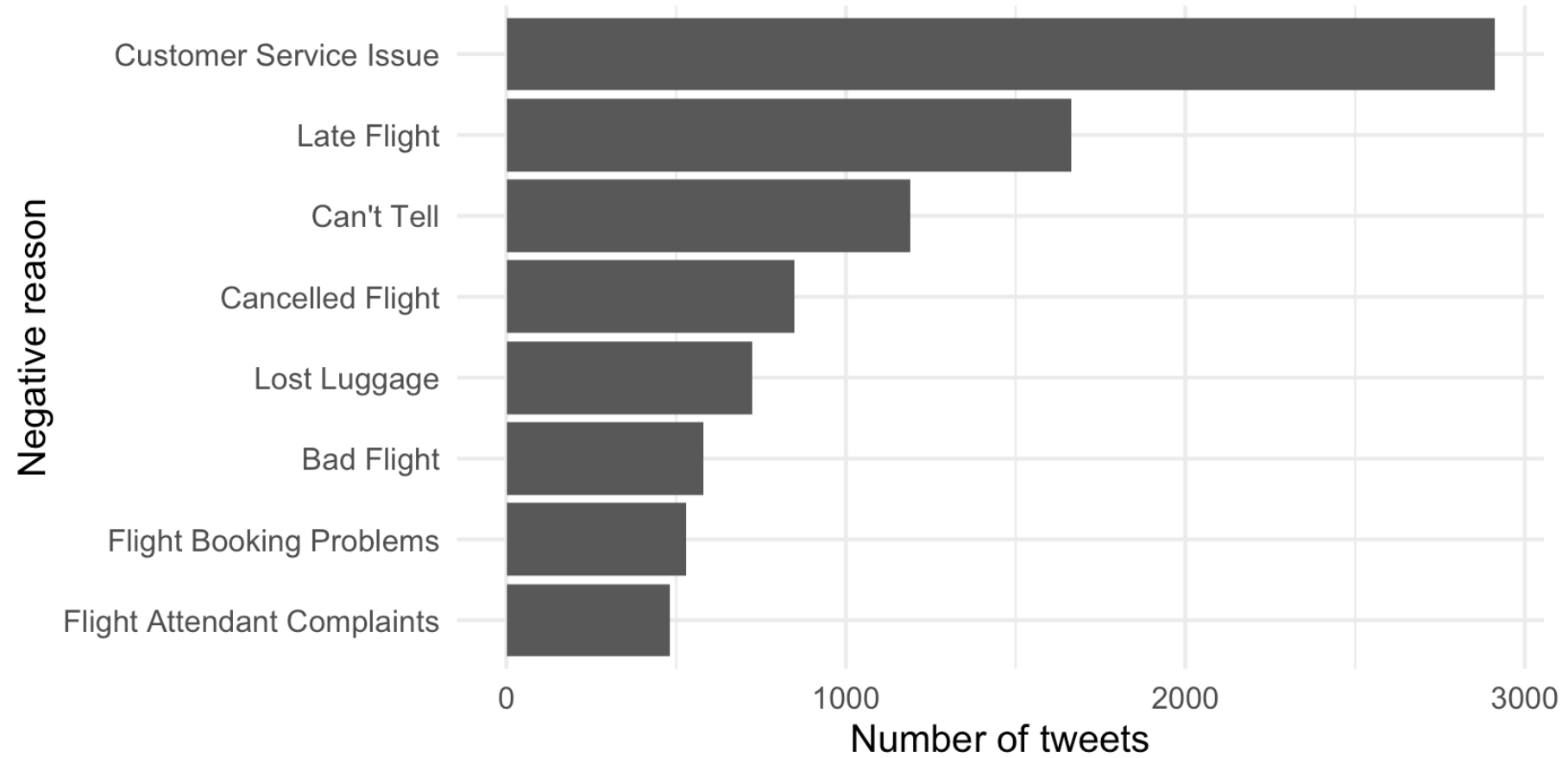
| Sentiment | Tweets | Share |
|-----------|--------|-------|
| negative  | 9178   | 62.7% |
| neutral   | 3099   | 21.2% |
| positive  | 2363   | 16.1% |

# Sentiment by Airline





# Main Negative Reasons



# R Workflow: Tokenization

```
1 tweets_words <- tweets |>
2   mutate(tweet_id = row_number()) |>
3   select(tweet_id, airline, airline_sentiment, text) |>
4   unnest_tokens(word, text) |>
5   anti_join(stop_words, by = "word")
```

## Example

*The flight was delayed and the service was terrible.*

## After tokenization:

flight delayed service terrible

# Lexicon-Based Sentiment Analysis

```
1 word_sentiments <- tweets_words |>  
2   inner_join(get_sentiments("bing"), by = "word")
```

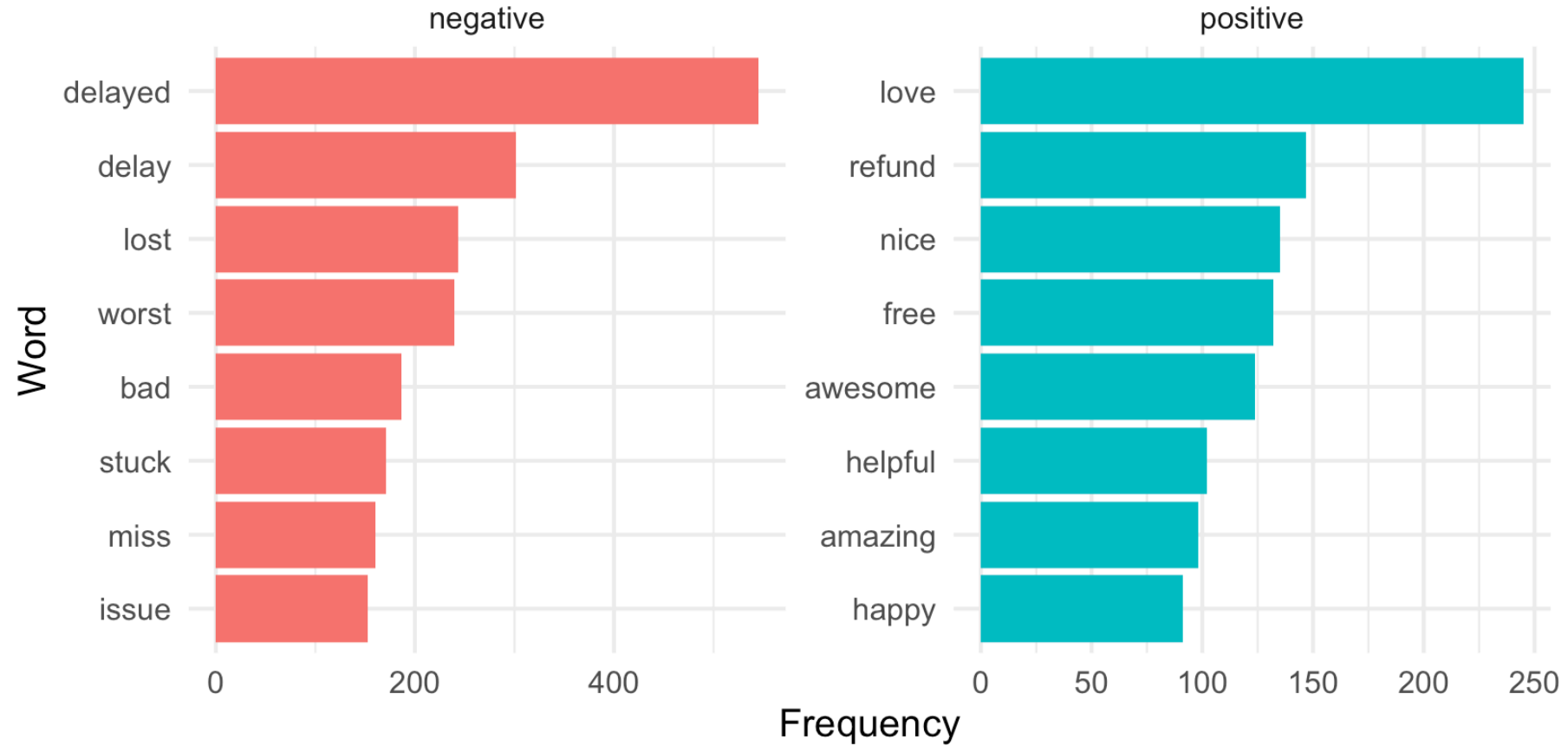
## Example word matching with Bing lexicon

*The flight was delayed and terrible, but the staff was helpful and comfortable.*

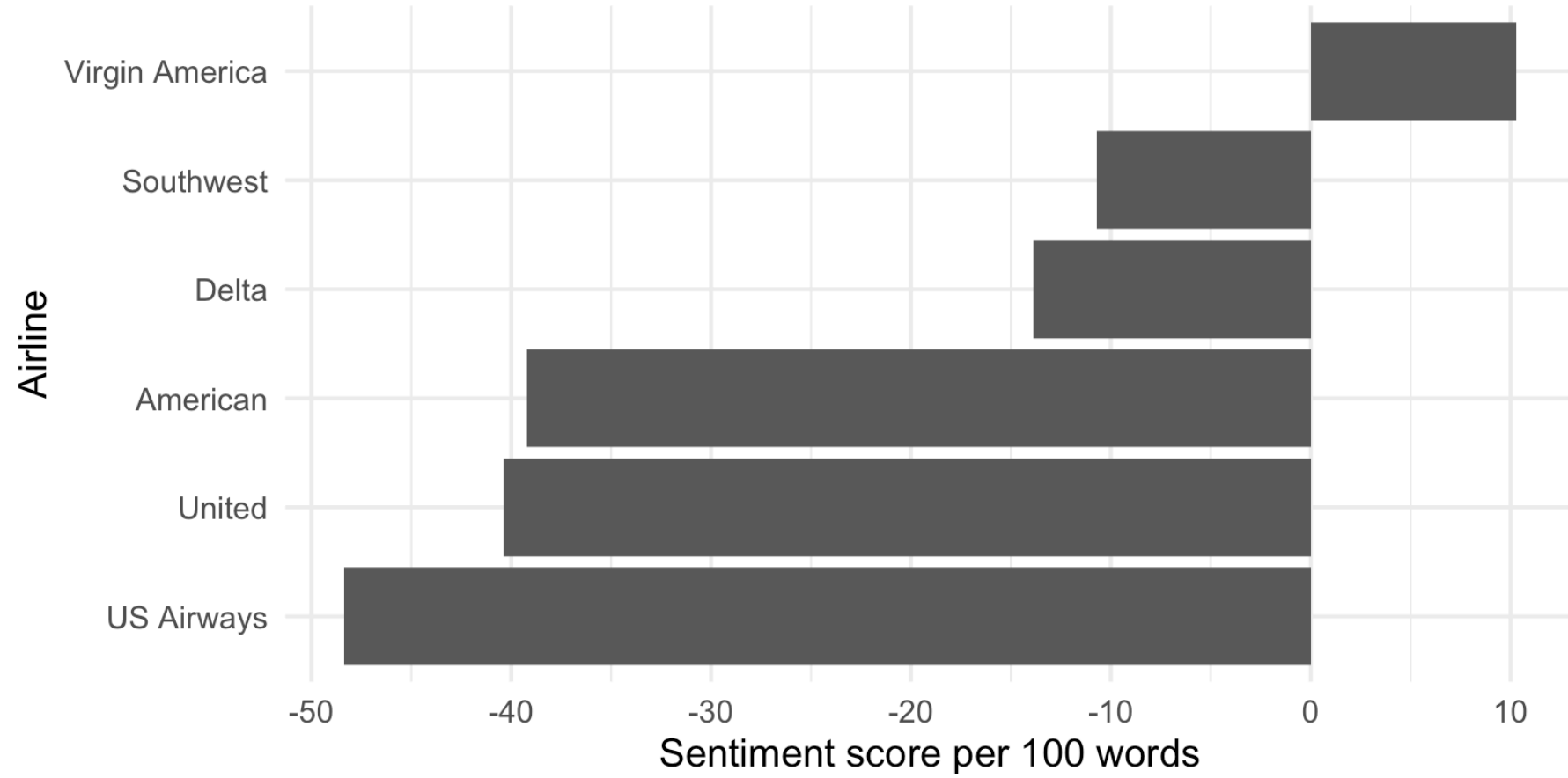
terrible → negative   delayed → negative   helpful → positive   comfortable → positive

Words that are not found in the dictionary are not labelled as neutral; they are excluded from the lexicon-based count.

# Frequent Sentiment Words



# Sentiment Score by Airline



# From Analysis to Decisions

| Problem                | Tweets | Managerial action                              |
|------------------------|--------|------------------------------------------------|
| Customer Service Issue | 2910   | Improve support capacity and service recovery. |
| Late Flight            | 1665   | Improve delay communication and punctuality.   |
| Can't Tell             | 1190   | Investigate this category further.             |
| Cancelled Flight       | 847    | Strengthen rebooking and compensation.         |
| Lost Luggage           | 724    | Improve baggage tracking.                      |

## **Managerial Interpretation**

**The real value is not only knowing whether customers are happy or unhappy.**

The real value is knowing which operational problem should be solved first.

# Strengths and Limitations

## Strengths

Fast analysis  
Reproducible workflow  
Useful for monitoring

## Limitations

Sarcasm is difficult  
Context can be missed  
Human interpretation is  
needed



# In-Class Exercise

Students analyze three short customer comments in R and recommend one managerial action.

```
1 customer_comments <- tibble(  
2   text = c(  
3     "The flight was delayed and the service was terrible",  
4     "The staff was helpful and the flight was comfortable",  
5     "My luggage was lost and nobody helped me"  
6   )  
7 )  
8  
9 customer_comments |>  
10  unnest_tokens(word, text) |>  
11  anti_join(stop_words, by = "word") |>  
12  inner_join(get_sentiments("bing"), by = "word") |>  
13  count(sentiment)  
14  
15 # Show which words were matched as positive or negative  
16 customer_comments |>  
17  unnest_tokens(word, text) |>  
18  anti_join(stop_words, by = "word") |>  
19  inner_join(get_sentiments("bing"), by = "word") |>  
20  select(word, sentiment) |>  
21  arrange(sentiment, word)
```

## Exercise Question

**Question:**

Which managerial action would be most suitable based on the negative words in the comments?

- A) Reduce the number of customer support channels
- B) Improve delay communication and luggage recovery processes
- C) Ignore negative comments because they are unstructured data
- D) Only focus on positive words

## Conclusion

**Sentiment analysis turns unstructured customer feedback into practical decision support.**

For managers, the key question is not only: **“What is the sentiment?”**

It is: **“What decision should this sentiment trigger?”**

# References

Huber, S. (2025). *How to use R for data science*. <https://hubchev.github.io/ds/>

Liu, B. (2012). *Sentiment analysis and opinion mining*. Morgan & Claypool Publishers. <https://www.cs.uic.edu/~liub/FBS/SentimentAnalysis-and-OpinionMining.pdf>

Pang, B., & Lee, L. (2008). Opinion mining and sentiment analysis. *Foundations and Trends in Information Retrieval*, 2(1–2), 1–135. <https://doi.org/10.1561/15000000011>

Silge, J., & Robinson, D. (2017). *Text mining with R: A tidy approach*. O'Reilly Media. <https://www.tidytextmining.com/>

Wickham, H., Cetinkaya-Rundel, M., & Grolemund, G. (2023). *R for data science: Import, tidy, transform, visualize, and model data* (2nd ed.). O'Reilly Media. <https://r4ds.hadley.nz/>

*Twitter US Airline Sentiment dataset*. Airline-related tweets used for sentiment and negative reason analysis. <https://raw.githubusercontent.com/ruchitgandhi/Twitter-Airline-Sentiment-Analysis/master/Tweets.csv>

# Thank You

**Mustafa Asım Ruhi**

Student number: 401190044

Data Analysis for Decision-Making

2 June 2026